MIT Professor Uses AI to Make Power-Hungry Data Centers More Efficient

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Huge buildings filled with computers, called **data centers**, use so much electricity that they can strain power grids and increase demand for energy from **fossil fuels**. Christina Delimitrou, a newly tenured associate professor at MIT, is trying to shrink that environmental cost by changing how servers and networks inside **data centers** work.

Her group uses **machine learning** to make large **cloud** systems more efficient, secure, and reliable. They redesign older **cloud** software, manage shared hardware more carefully, and design leaner servers so operators can get more computing done with the machines they already have.

"If **data centers** are not utilized to the best of their capabilities, then they will burn much more power than they need to meet growing user demand," Delimitrou said. She noted that software often includes extra "bloating," and removing it without hurting performance could mean fewer new **data centers** need to be built.

Her team also uses AI to help programmers find and fix problems in **cloud** apps such as music streaming or video calls, which can cut downtime that wastes computing power. Better resource management, she said, can give phone users more predictable performance.

Earlier research with mentor Christos Kozyrakis found many large systems ran at only about 15 percent **capacity** even when demand was high. **Machine learning** can automate resource choices that are hard for people to manage at the scale of tens of thousands of machines.

Delimitrou grew up in northern Greece, studied computer engineering in Athens, earned a Ph.D. at Stanford, taught at Cornell, and now works in MIT's Department of Electrical Engineering and Computer Science and **CSAIL**. Her work shows how smarter softwarenot only more buildingscan answer growing demand for **cloud** computing.

Word Treasure

data centers -- Large buildings packed with computers that store and process online information.

machine learning -- A kind of artificial intelligence that finds patterns in data to improve decisions.

cloud -- Remote computing services delivered over the internet.

fossil fuels -- Energy sources such as coal, oil, and gas formed from ancient plants and animals.

capacity -- How much work machines could do when fully used.

CSAIL -- MIT's Computer Science and Artificial Intelligence Laboratory.

Background you need

Data centers are large buildings filled with computers that power websites, apps, and cloud services.

Those machines need electricity for computing and cooling, so inefficient systems can waste a lot of power.

Machine learning can help decide how to share servers and networks more wisely.

Using existing hardware better can reduce the need to build as many new facilities.

Break it down

LEAD: MIT professor uses AI to cut wasted power in data centers
KEY EVENT: Research redesigns cloud servers, networks, and resource sharing
+- WHO: Christina Delimitrou and her MIT/CSAIL group
+- WHY: Growing data centers strain grids and fossil-fuel demand
EVIDENCE: Earlier finding of ~15% utilization; machine learning to automate management
IMPACT: Better phone-app performance and possibly fewer new data centers
OPEN QUESTION: How widely operators will adopt these efficiency methods

Why it matters

Cloud apps kids use every day run in data centers. Smarter software can mean less wasted electricity and a smaller environmental footprint.

Test what you remember

1. Where does Christina Delimitrou work?

(A) A bakery
(B) The U.S. Mint
(C) MIT
(D) NASA Headquarters only

2. What problem do growing data centers create?

(A) They ban smartphones
(B) They use lots of electricity and can strain grids
(C) They erase all books
(D) They stop rain from falling

3. What tool does her group apply to improve efficiency?

(A) Horse-drawn carts
(B) Machine learning
(C) Handwritten letters only
(D) Paper airplanes

4. About what capacity did earlier research find many systems running?

(A) 200 percent
(B) About 15 percent
(C) Exactly 100 percent
(D) Zero percent forever

5. What does Delimitrou say about software "bloating"?

(A) It makes computers colder
(B) It only helps video games
(C) Removing it without hurting performance could mean fewer new data centers
(D) It is required by law

6. Besides efficiency, what else can AI help programmers do in this story?

(A) Find and fix problems in cloud apps
(B) Build rockets on the Moon
(C) Write every newspaper
(D) Grow olive trees

Answer key (try the quiz first!): 1: C 2: B 3: B 4: B 5: C 6: A

Questions to think about

1. Christina Delimitrou: Argues underused data centers burn extra power and that removing software bloat can reduce the need for new facilities.

MY THOUGHTS (at least 20 words):